PyGameMaker — Tutorial 2: Your First Game — Catch the Star

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Your First Game: Catch the Star** (5 pages, about 15-20 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase before moving on. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A complete little game: you move a spaceship left and right along the bottom of the screen, stars fall from the sky, and every star you catch gives you 10 points.

Phase 1: A Moving Player

- ☐ **1.** Make a new project called CatchTheStar (**File > New Project**).
- ☐ **2.** Create a sprite spr_player (32×32) and draw a spaceship, a person or a basket. Save with **Ctrl+S**.
- ☐ **3.** Create an object obj_player and give it the sprite spr_player.
- ☐ **4.** Create a room room_game and place obj_player near the bottom centre.
- ☐ **5.** In obj_player, open the **Blockly** tab and add three events: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -5*, **Keyboard: Right Arrow (held)** with *Set horizontal speed to 5*, and **Keyboard: No key** with *Stop movement*.

> **You should see:** press **F5**. The arrow keys move your player left and right, and it stops when you let go. It can leave the screen — that is expected for now.

Phase 2: Falling Stars

- ☐ **6.** Create a sprite spr_star (32×32) and an object obj_star that uses it.
- ☐ **7.** In obj_star, add a **Create** event with *Set vertical speed to 3*.
- ☐ **8.** Create an object obj_spawner with **no sprite**. Its **Create** event does *Set Alarm 0 to 60*. Its **Alarm 0** event does *Create instance of obj_star at x: random, y: 0* and then *Set Alarm 0 to 60* again.
- ☐ **9.** Place obj_spawner anywhere in room_game.

> **You should see:** press **F5**. About one star per second appears at the top, in a different place each time, and falls. Stars pass straight through your player — that is expected for now.

Phase 3: Catching and Scoring

- ☐ **10.** In obj_player, add **Collision with obj_star**: *Add to score 10*, then *Destroy other instance*.
- ☐ **11.** Create obj_game_controller (no sprite). Its **Create** event does *Set score to 0*; its **Draw** event does *Draw score at x: 10, y: 10*.
- ☐ **12.** In obj_star, add an **Outside Room** event with *Destroy this instance*.

- [ ] **13.** Place obj_game_controller anywhere in the room.

> **You should see:** press **F5**. Catch a star: it disappears and the score at the top left goes up by 10. Stars you miss disappear at the bottom.

Phase 4: Finishing Touches

- [ ] **14.** In obj_player, add a **Step** event that keeps the player on the screen: if x is less than 0, set x to 0; if x is greater than the room width minus the sprite width, set x to that value.
- [ ] **15.** Give room_game a background colour you like.

> **You should see:** the player stops at both edges of the window, and your finished game is ready to play.

Stuck?

What is wrong	Most likely cause	What to do
The player does not move	The events are on the wrong object, or obj_player is not in the room	Open obj_player and check the three Keyboard events; open the room and check the player is placed
The player keeps sliding after I let go	The **No key** event is missing	Add **Keyboard: No key** with *Stop movement*
No stars appear	obj_spawner is not in the room, or its Create event does not set the alarm	Place the spawner; check *Set Alarm 0 to 60* is in its **Create** event
Only one star ever appears	The alarm is not set again at the end of **Alarm 0**	Add *Set Alarm 0 to 60* as the last action of **Alarm 0**
The score does not show	obj_game_controller is not in the room, or it has no Draw event	Place it; add *Draw score* in a **Draw** event
The score never goes up	The Collision event is on the wrong object or has no *Add to score*	Put **Collision with obj_star** on obj_player
My player disappears when it touches a star	I used *Destroy this instance* instead of *Destroy other instance*	Use **Destroy other instance**

Challenges

- **Try this (5 minutes):** change the star's fall speed, or the number of points for a star.
- **Push further:** make stars fall faster as the score goes up, or add a second kind of star worth more points.
- **Invent:** add a "bomb" object that ends the game if you catch it, or lives that you lose when a star reaches the bottom.

Vocabulary

Term	What it means
Event	Something that happens in the game (a key is pressed, two objects touch)
Action	What an object does when an event happens
Alarm	A timer; it counts down and then triggers its Alarm event
Spawner	An invisible object that creates other objects
Collision	When two objects touch

Check Yourself

Write your answers in the notes below.

1. What does an **alarm** do, and why does the spawner set it again at the end?
2. What is the difference between *Destroy this instance* and *Destroy other instance* in the collision event?
3. Which object draws the score, and why is it not the player?

My Notes